Veyant Technical Architecture

Agentic GraphQL Mesh for Travel Orchestration

Moving from sequential MCP tool calls to parallel GraphQL queries reduces latency from 6+ roundtrips to 1 roundtrip

PLATFORM
Azure PaaS

AI ENGINE
Anthropic Claude

PATTERN
GraphQL Mesh

Brain + Hand + Muscle

AI reasoning separated from data orchestration

Why Traditional MCP Falls Short

TRADITIONAL MCP PATTERN

1. Call GDS for flights - 800ms
2. Get traveler preferences - 200ms
3. Check corporate policy - 150ms
4. Query hotel inventory - 600ms
5. Get loyalty status - 300ms
6. Check car rental - 400ms

Total: 2.5 seconds (sequential)

AGENTIC MESH PATTERN

Single GraphQL query requests all data

Gateway executes in parallel:

GDS, Hotels, Cars run simultaneously

Traveler context loaded in parallel

Policy checked alongside inventory

Total: 800ms (parallel)

Key Insight: The AI describes WHAT data it needs, the gateway handles HOW to fetch it.

Brain, Hand, Muscle Architecture

BRAIN

AI Reasoning Layer

Claude (Anthropic API)

- Intent understanding
- Query composition
- Decision making
- Response synthesis

Azure: AI Services / Direct API

HAND

Orchestration Gateway

GraphQL Mesh Gateway

- Query execution
- Parallel data fetching
- Schema federation
- Field-level security

Azure: API Management + Functions

MUSCLE

Backend Systems

Data Sources + APIs

- GDS/NDC APIs
- Hotel/Car systems
- Travel Cube (Cosmos)
- Policy database

Azure: Cosmos DB + Connectors

User Request → Brain (Claude) → GraphQL Query → Hand (Gateway) → Muscle (APIs) → Response

Azure PaaS Implementation

AI LAYER (Brain)

Anthropic Claude API (direct HTTPS) | Azure AI Services (embeddings, vision) | Blob Storage (prompt versioning)

ORCHESTRATION LAYER (Hand)

Azure API Management (GraphQL gateway, rate limiting) | Azure Functions (resolvers) | Event Grid (async workflows)

DATA LAYER (Muscle - Storage)

Cosmos DB (Travel Cube, tenant data) | Key Vault (credentials, HSM) | Redis Cache (session, availability)

CONNECTIVITY LAYER (Muscle - External)

GDS/NDC Connectors (Sabre, Amadeus, United) | Hotel/Car APIs | TMC Systems (back/mid-office)

Security

Azure AD + Managed Identity + Private Endpoints

Observability

Azure Monitor + Log Analytics + App Insights

DevOps

Bicep/Terraform + GitHub Actions

Three Cube Schema Federation

My Travel Cube

type Traveler { id, preferences, loyalty, credentials, tripHistory }

Azure: Cosmos DB (partition: tenantId+travelerId)

Corporate Cube

type CorporatePolicy { companyId, rules, preferredSuppliers, rates }

Azure: Cosmos DB (partition: companyId)

Supplier Cube

type SupplierInventory { flights, hotels, cars, availability }

Azure: Functions (resolvers to GDS/NDC/APIs)

Unified GraphQL Schema

```
type Query { searchTrip(input): TripResult, getTravelerContext(id): Context, checkPolicy(booking):
checkPolicy(booking): PolicyResult }
```

Single query fetches from all cubes in parallel

Federation Benefits

Single Endpoint - AI writes one query for all data

Parallel Fetch - Gateway optimizes execution

Field Security - Per-field authorization

Field-Level Security Architecture

THREE-TIER DATA CLASSIFICATION

Tier 1: Critical (HSM Protected)

Payment cards, passwords, supplier credentials

Azure: Key Vault Premium + Managed HSM

Tier 2: Sensitive (Encrypted)

Passport, loyalty accounts, traveler identity

Azure: Cosmos DB encryption + row-level security

Tier 3: Operational (Standard)

Preferences, trip history, search patterns

Azure: Cosmos DB with tenant isolation

GRAPHQL FIELD AUTHORIZATION

Schema-Level Access Control

```
type Traveler {
 id: ID!
 preferences: Preferences
 paymentMethods: [Card] @auth(tier:1)
 passport: Passport @auth(tier:2)
 tripHistory: [Trip] @auth(tier:3)
 }
```

Key Advantage: Security enforced at schema level, not in prompts. The AI cannot request data it should not access.

Token Efficiency: Cost at Scale

TRADITIONAL MCP: CONTEXT BLOAT

Each tool call returns full JSON to context:

Flight search results: ~8,000 tokens

Hotel availability: ~5,000 tokens

Car rental options: ~3,000 tokens

Traveler context: ~2,000 tokens

Policy rules: ~1,500 tokens

Total: ~19,500 tokens/query ($0.06-0.15)

GRAPHQL MESH: LEAN CONTEXT

Gateway joins data, returns only what is needed:

GraphQL query: ~300 tokens

Filtered flight results: ~1,200 tokens

Top hotel matches: ~800 tokens

Relevant traveler prefs: ~400 tokens

Applicable policies: ~300 tokens

Total: ~3,000 tokens/query ($0.009-0.023)

6.5x
Token Reduction

85%
Cost Savings

3x
Faster Response

1
Roundtrip vs 6+

Implementation Roadmap

Phase 1: Prototype
Dec 2025

AI: Claude API direct
Gateway: Simple REST
Data: Cosmos + stubs

Est: $280/mo Azure

Phase 2: Pilot
Q1 2026

AI: Azure AI + Claude
Gateway: APIM + GraphQL
Data: Live connectors

Est: $840/mo Azure

Phase 3: Production
Q2-Q3 2026

AI: Full mesh orchestration
Gateway: Schema federation
Data: Multi-tenant

Est: $3-5K/mo Azure

Phase 4: Scale
Q4 2026+

AI: Multi-model, fine-tuned
Gateway: Partner API platform
Data: Global distribution

Usage-based scaling

Architecture Decision: Start simple (Phase 1-2), but design for GraphQL mesh from day one. Schema-first approach enables smooth evolution.